

Pancreatic Cancer Prognosis using Clinical and Radiomic Data

UE24CS352A Machine Learning · Mini-Project · Problem #6

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• Problem Statement

Most pancreatic cancer patients are diagnosed late, so prognosis matters for borderline-resectable and locally advanced cases. We build survival-analysis models that predict patient risk from **clinical** features, **radiomic** (MRI-derived) features, and both combined, and test whether radiomics adds prognostic value. The task follows the reference report by A. Jamalain (Stanford CS229, 2020).

• Dataset

The reference cohort (80 patients, 900 QIFP radiomic features, plus clinical fields such as stage, KPS, resectability, chemotherapy and tumour size) was provided by Stanford Cancer Center under an NDA and is **not public**. We therefore generated a **synthetic cohort with the same structure** (src/generate_data.py): 80 patients, 12 clinical features, 900 correlated radiomic features (30 latent factors), right-censored survival times (57 events, median 336 days) drawn from a hidden Cox-style hazard.

Results are illustrative of the pipeline, not clinical findings, and are not directly comparable to the paper's numbers.

• Approach

- (i) **Risk-threshold selection:** Naive Bayes (Laplace-smoothed, discretised clinical features) classifies high/low risk for survival-day thresholds 240-360; balanced accuracy under 10-fold CV picks the threshold.
- (ii) **Augmentation:** SMOTE oversamples the small high-risk group (time and event are interpolated with the features).
- (iii) **Cox proportional hazards** with L2 penalty (0.1, l1_ratio=0) on clinical (12), radiomic (150 highest-variance) and combined feature sets; radiomic features with a 95% CI width on log(HR) above 1 are dropped to cut variance.
- (iv) **Evaluation:** 10-fold CV Harrell C-index; Kaplan-Meier curves for patients split at the median hazard, with a log-rank test.
- (v) **Improvement over the paper:** SMOTE, standardisation and feature filtering are fitted *inside each training fold only*, avoiding leakage into held-out folds.

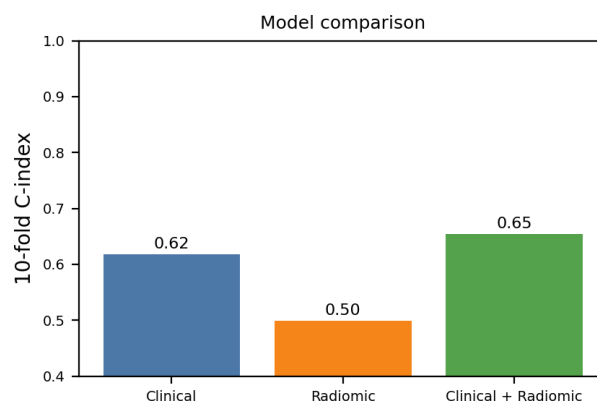
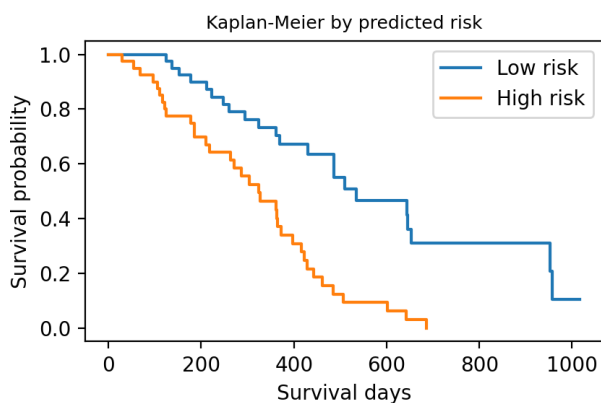
• Implementation Overview

Python 3 with *lifelines* (Cox, KM, log-rank), *imbalanced-learn* (SMOTE), *scikit-learn* (Naive Bayes, CV) and *matplotlib*. generate_data.py builds the cohort; pipeline.py runs all stages and saves metrics and figures to results/; demo.py scores a patient for the live demo. Everything runs with python src/pipeline.py (see README).

• Results

Model	Features in	Used (avg/fold)	10-fold C-index
Clinical	12	12	0.618 ± 0.133
Radiomic	150	150	0.499 ± 0.165
Clinical + Radiomic	162	7	0.654 ± 0.142

Naive Bayes balanced accuracy by threshold: 240 d: 0.46, 270 d: 0.50, 300 d: 0.33, 330 d: 0.41, 360 d: 0.42. The best threshold was **270 days**, but accuracy stays near chance, so with 80 patients the threshold choice is weakly supported. The log-rank test between predicted high- and low-risk groups gave a statistic of 18.9 ($p = 1.3e-05$); this final-model test is computed in-sample and is therefore optimistic.



• Conclusions

The full pipeline works end to end: Naive Bayes threshold selection, SMOTE, penalised Cox regression and risk stratification. Combining clinical and radiomic features (C-index 0.65) beat clinical alone (0.62), consistent in direction with the paper, while the radiomic-only model was near chance because 150 correlated features overfit ~70 training patients. Fold variance is large (± 0.13 -0.17), so these differences are not conclusive.

Limitations: synthetic data, tiny sample, and synthetic survival times from SMOTE interpolation.

Future work: real data, stronger regularisation or PCA on radiomics, and DeepSurv/DeepHit models.

Reference: A. Jamalian, *Pancreatic cancer prognosis using clinical and radiomic data*, CS229 Final Report, Stanford, 2020. <https://cs229.stanford.edu/proj2020spr/report/Jamalian.pdf>